

STAR-Pipeline: Spacecraft Telemetry Anomaly Detection & Resolution

Fine-Tuning Open LLMs for Mission-Critical Infrastructure



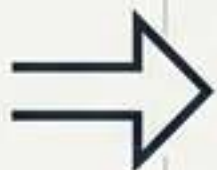
A rigorous empirical bake-off: Fusing classical LSTMs with localized LLMs out-detects frontier APIs while delivering sovereign, human-readable diagnostic advice.

The Mission-Critical Dilemma: Sovereignty vs. Capability

Can a localized, open-source model running on-premise match traditional machine learning detectors and beat frontier cloud APIs?

The Requirement

[0.61, 0.58, 0.60,
0.60, 0.62, 0.59,
0.63, 0.57, 0.61]



DIAGNOSIS:
Thermal drift.
ACTION: Inspect
calibration.



Detect anomalies and tell operators exactly what to do.

The Constraints

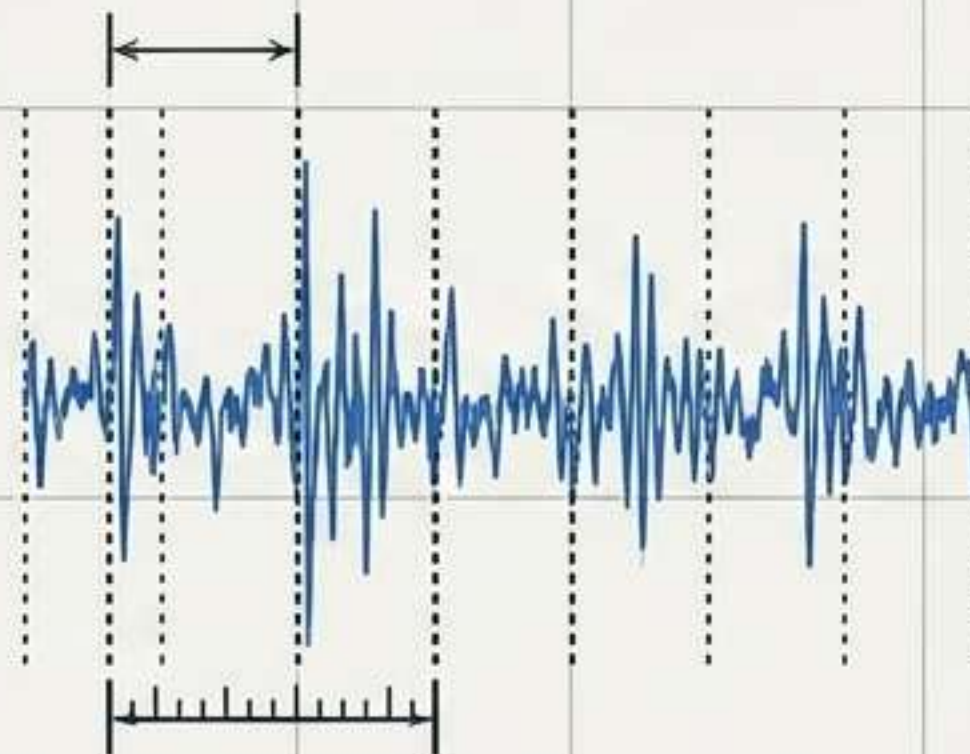
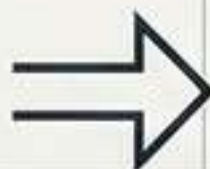


No external API dependencies (sovereignty),
high cost of false alarms (alert fatigue),
highly specialized data.

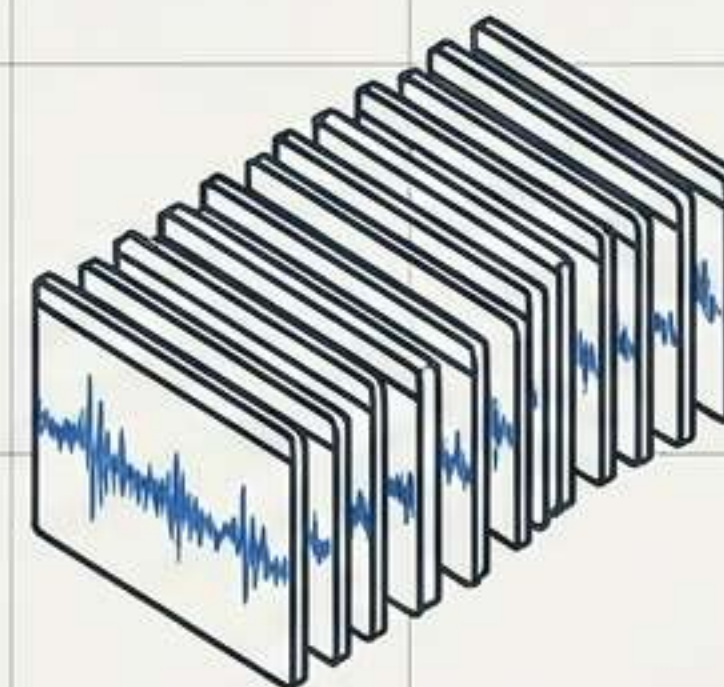
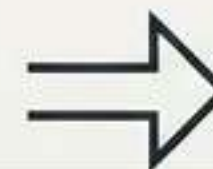
Chopping the Infinite Stream: The ESA-AD Dataset



29 GB of raw telemetry
(3 ESA missions, 224 channels)



32-step sliding windows
1-hour cadence



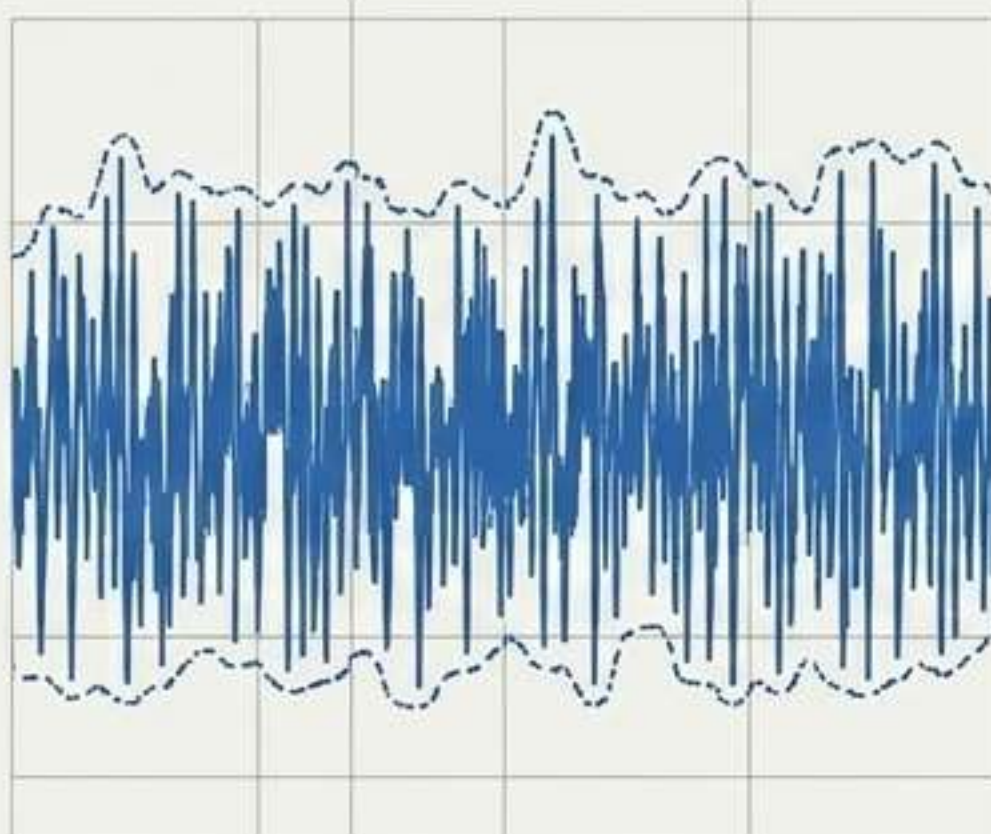
RevIN Normalized
Patches

The Final Exam Split

4,500-window test set. Deliberately balanced to 25% **anomalies** for an honest, interpretable score.

The Contenders: Three Approaches to Anomaly Detection

Classical Gatekeeper LSTM (Telemanom-style)



An autoencoder that checks for reconstruction errors.
Fast, classical machine learning.

The Text Advisor Qwen3-8B (QLoRA)

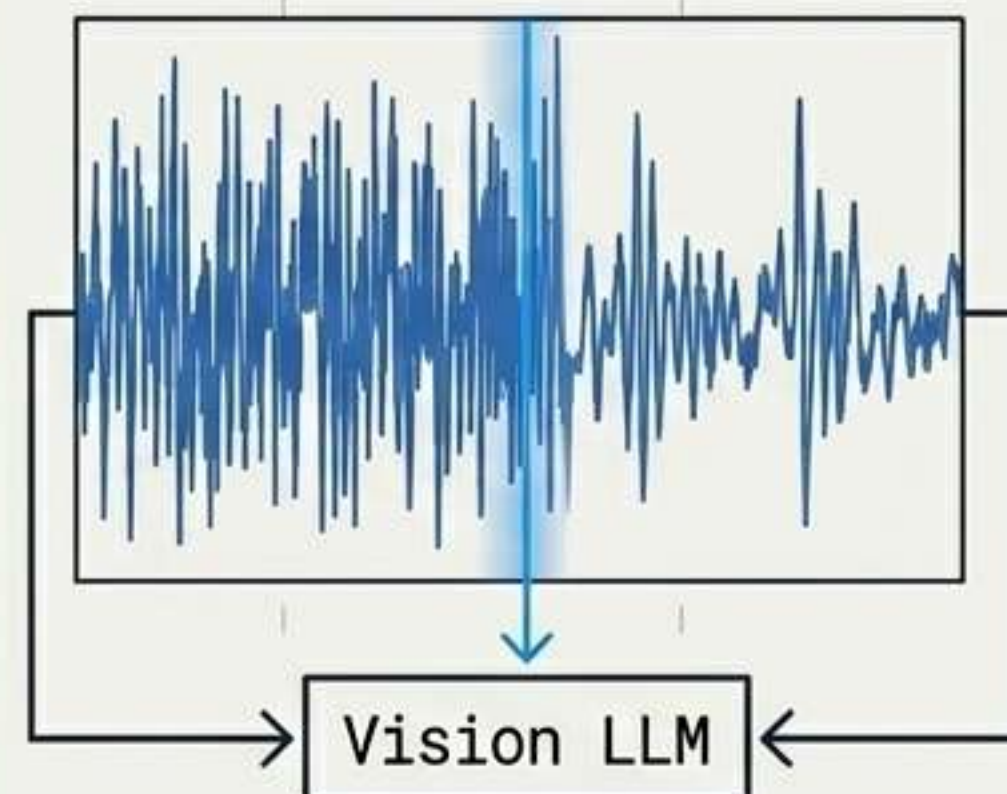
```
[0.61, 0.58, 0.60, 0.60, 0.62, 0.59,  
0.60, 0.62, 0.59, 0.63, 0.57, 0.61,  
0.60, 0.62, 0.59, 0.63, 0.57, 0.68,  
0.60, 0.62, 0.59, 0.63, 0.57, 0.61]
```



DIAGNOSIS: Thermal drift detected in sensor channel A4. Potential calibration issue.
ACTION: Inspect calibration and verify sensor integrity. Monitor for further deviation.

Reads telemetry values as raw text. Generates both anomaly verdicts and human-readable diagnostic advice.

The Vision Inspector Qwen3-VL-8B



A multimodal vision model fine-tuned on rendered PNG plots of the telemetry. Highly precision-oriented.

Setting the Honest Bar: The Floor of Competence

The Trap

If a model blindly flags every window as an anomaly on this dataset, it gets an F1 score of ~ 0.40 for free.

The Rule

Any approach scoring below 0.40 hasn't learned to detect; it is only guessing.

Detection

Always-Anomaly Baseline (F1: 0.399)

Just Over-Flagging

The Ultimate Metric: CEF0.5

A precision-weighted F-beta score, because in a control room, a false alarm is far more costly than a near-miss.

Question 1: Does Fine-Tuning Actually Help?

Fine-tuning teaches the model to obey the format contract and provides the domain discrimination that APIs lack.

	Zero-Shot / Few-Shot (Prompting)	Fine-Tuned (SFT)
Base Model (Qwen3-8B)	Zero-Shot: F1 0.000 (Fails format) Few-Shot: F1 0.420 (Blindly flags 80%)	F1 0.453 The only LLM to clear the floor with a balanced precision/recall.
Frontier Model (Claude API)	F1 0.254 / 0.239 Sits at chance, completely below the competence floor.	N/A Frontier APIs lack local fine-tuning capabilities.

0.399 Trivial Baseline

The Channel History Deficit: Why Prompting Fails

What the Prompt Sees

[0.61, 0.58, 0.60, 0.59, 0.62, 0.61, 0.60]



Context-Free

A reading of 0.60 is normal for a voltage line, but critical for a thermal line.

Without history, there is no signal.

What Fine-Tuning Learned

[0.61, 0.58, 0.60, 0.59, 0.62, 0.61, 0.60]

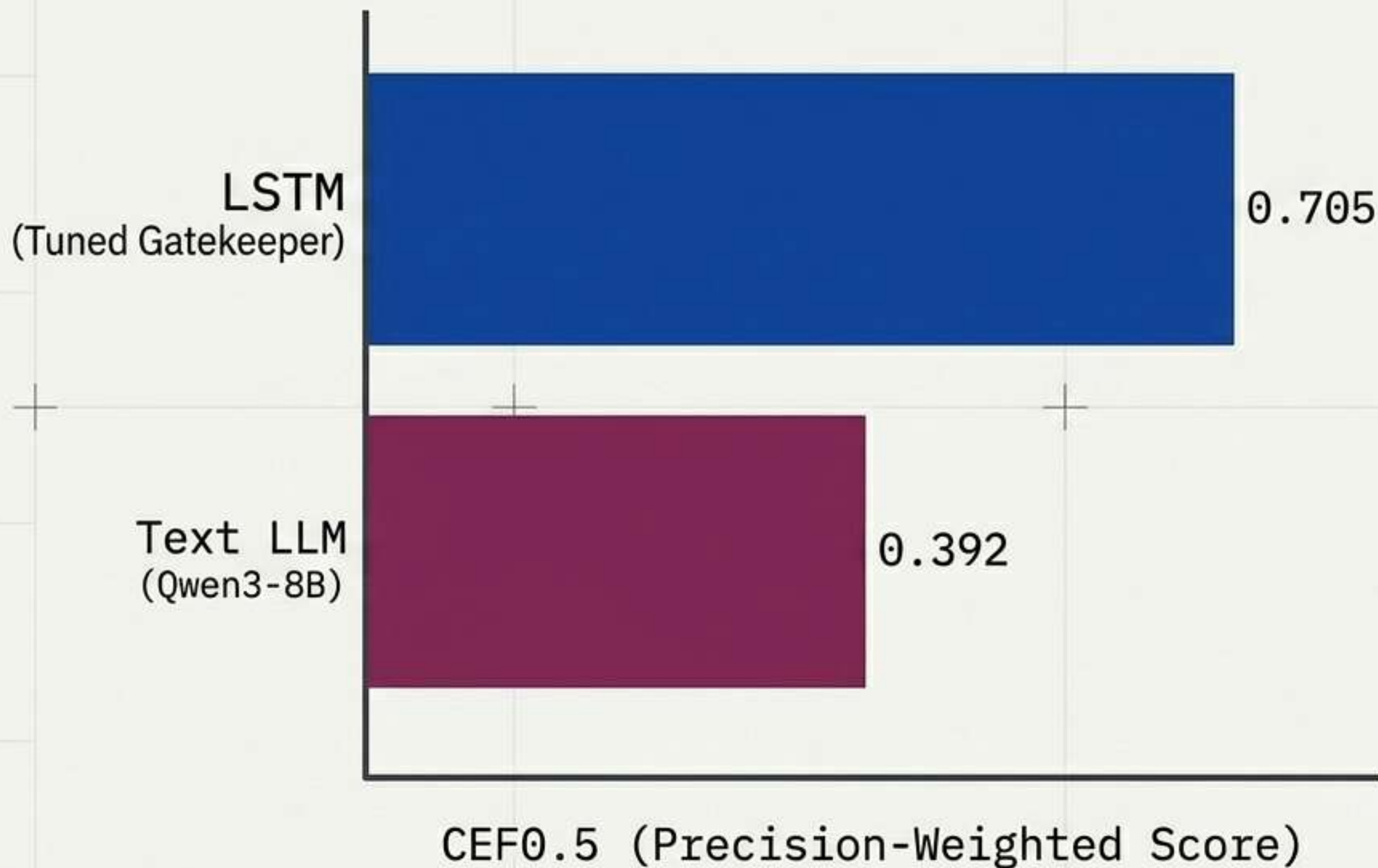
Mission1/channel_41

Localized Priors

Frontier models have never seen this specific spacecraft's telemetry.

Fine-tuning bakes 21,000 windows of localized "normal" patterns directly into the model weights.

Question 2: Can a Single LLM Out-Detect Classical ML?



The Verdict

No. A single LLM does not out-detect a tuned classical LSTM.

The LLM Weakness

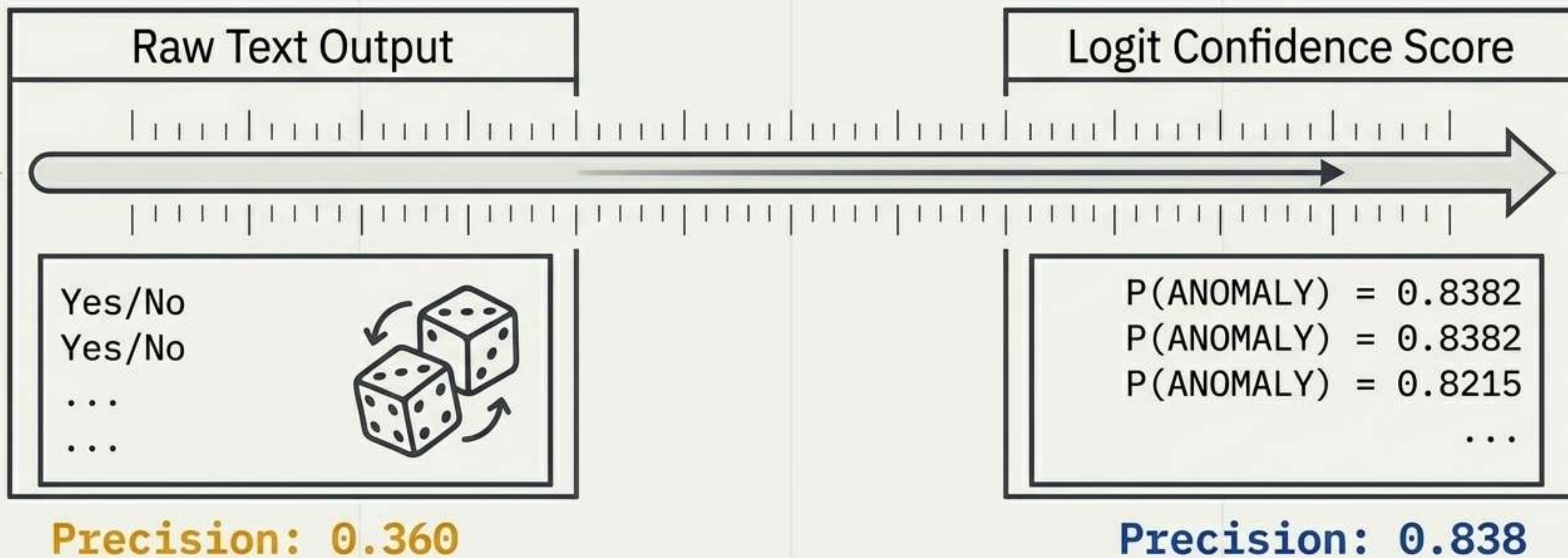
Out of the box, the Text LLM trades precision for recall. It over-flags anomalies (Precision 0.360).

The Silver Lining

LLMs can do something the LSTM cannot—generate plain-English diagnostic advice.

The Calibration Twist: Fixing the LLM's 'Over-Flagging'

The text model wasn't a bad detector; we were reading it badly. By eliminating text-sampling randomness and reading the continuous confidence score, precision more than doubled.



The Missing Link: Vision as the Counterweight

Because the Text and Vision models process the data through completely different modalities, they make independent errors. This makes them perfect candidates for mathematical fusion.

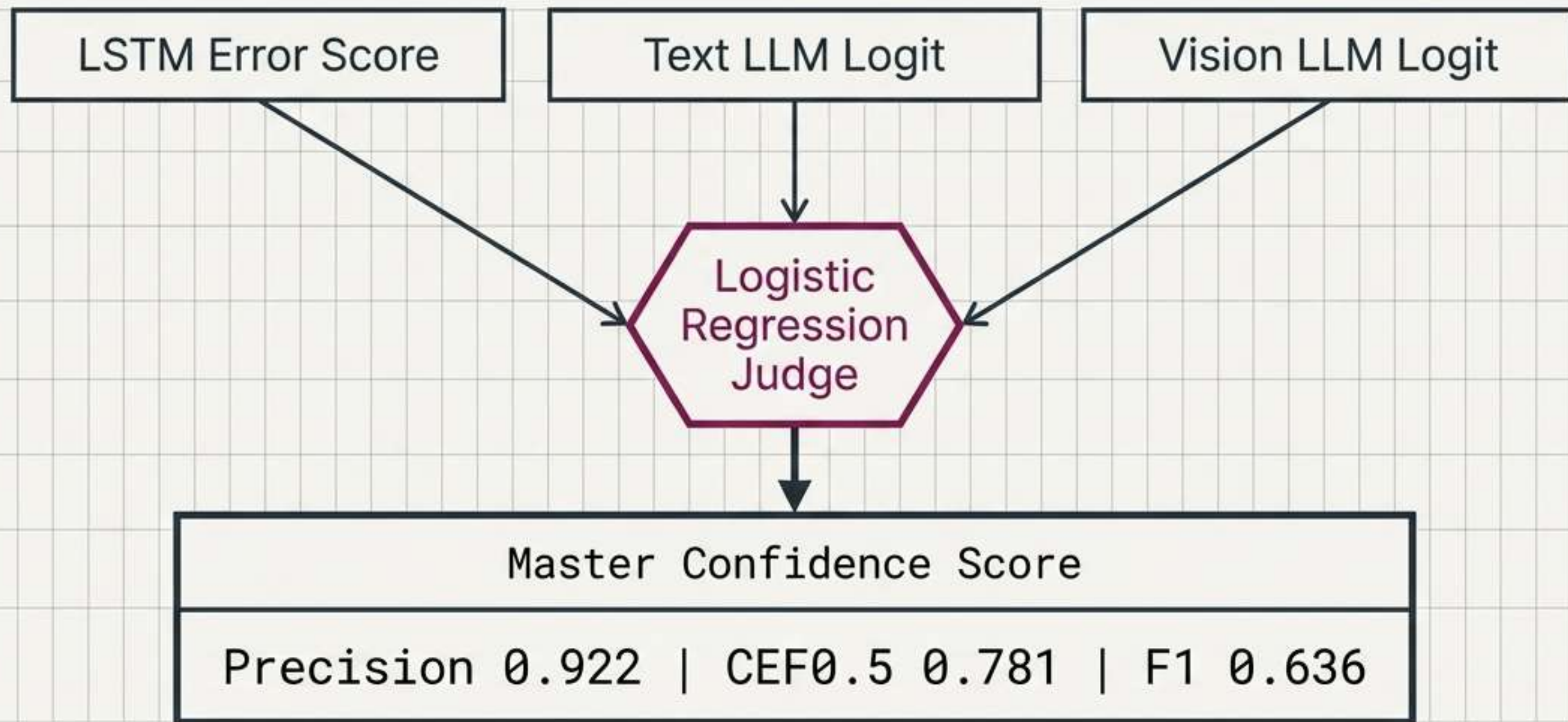
Text LLM	
[Recall-Oriented]	
Precision: 0.360	Recall: 0.609
Finds anomalies easily, but prone to false alarms. Reads raw numeric arrays.	

+

Vision LLM	
[Precision-Oriented]	
Precision: 0.769	Recall: 0.325
Very conservative, almost never false-alarms. Fine-tuned on rendered PNG plots of telemetry.	

The Synthesis: Fusing Independent Errors

Fusing models that make independent mistakes mathematically guarantees a Pareto-dominant detector. The ensemble beats every single individual model.



The Advice Layer: Quality & Grounding

The LLM writes highly accurate, grounded advice—but only when the premise is correct. Advice generation must be strictly gated by the high-precision ensemble detector.

On True Positives:	95% High-Quality (5.58/6 score)
Grounding:	119/120 times, names correct channel. 117/120 times, correct subsystem.
On False Positives:	Score drops to ~1/6 . The model confidently hallucinates an explanation for a non-existent anomaly.

VERDICT: **ANOMALY**

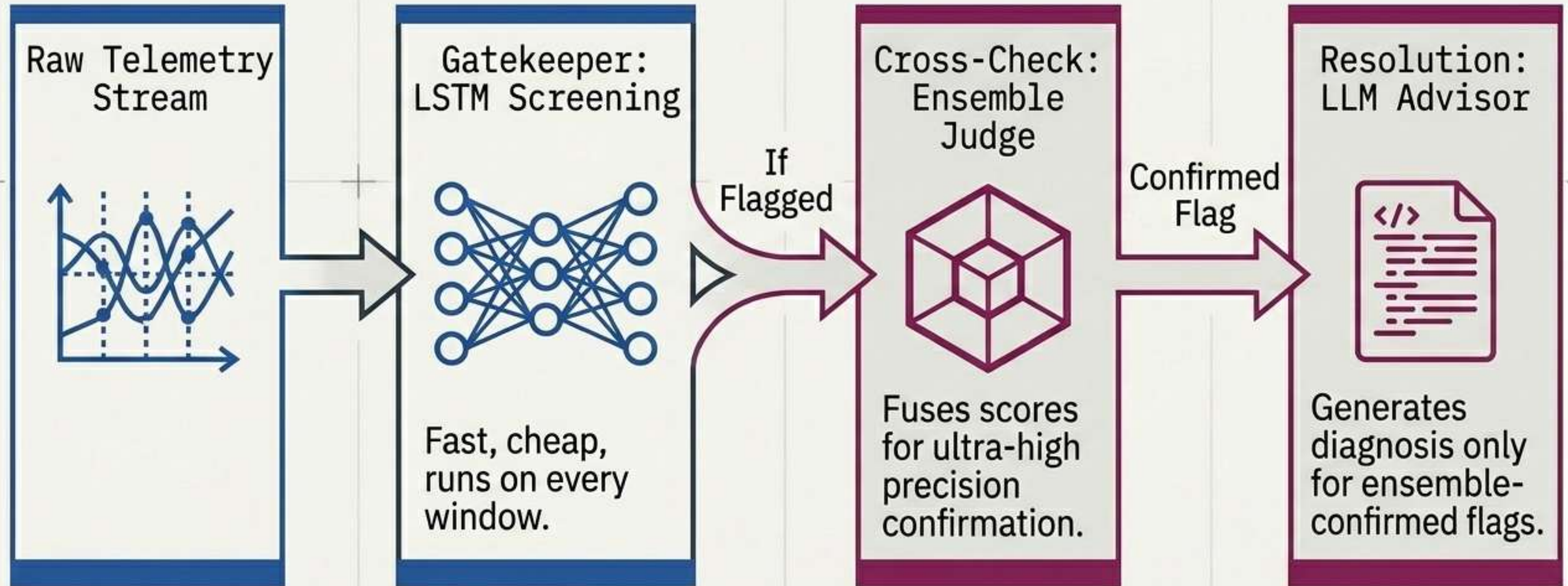
DIAGNOSIS: Thermal drift in payload subsystem.

ADVICE: Verify reference baseline against housekeeping archive.

ACTION: Schedule inspection at next ground contact.

The Final Architecture: The Hybrid Blueprint

Cheap where it can be, precise where it matters, and explainable at the end.
All running locally with zero external API dependency.



The Bottom Line

	Approach	Precision	Advice Capability	Inference Cost
1	Trivial Baseline	0.250	None	Free
2	LSTM	0.837	None	Instant
3	Text LLM	0.360	Yes	2.77s
4	Hybrid Ensemble	0.922	Yes	On-Demand

Proof: Fine-tuning localized open models defeats generic frontier prompting.

Reality: LLMs shouldn't replace classical detectors; they should fuse with them and explain them.

Status: A rigorous, reproducible framework ready for sovereign, mission-critical deployment.